



Baggage Handling Best Practices

An Analytical Report Generated by the AI Research Council

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Abstract

Baggage handling system (BHS) failures are the most predictable operational disaster in aviation and the least learned-from. Every major BHS embarrassment of the last 25 years — DEN '95, HKG '98, T5 '08, BER '20 — traced to the same small set of causes: integrated testing compressed or deferred, optimistic throughput assumptions, and an owner unwilling to delay opening for a system that wasn't ready. The state of the art in BHS is now mature enough that the technology is rarely the failure mode; governance, testing discipline, and operational integration are. MWAA's BHS investments at IAD and DCA should be evaluated against this reality, not against vendor performance specs. The report examines this claim across the following: The actual failure taxonomy of major BHS deployments since 1995. What broke, when did the warning signs appear, and what did the owner do or fail to do?; ICS (Independent Carrier System) vs. integrated DCV vs. tilt-tray vs. cross-belt: where does each actually win? Tradeoffs as a function of terminal geometry, peak-hour bag volume, and connection complexity. Not a vendor comparison — an architectural one; The 95% rule and what it hides: stated read rates, jam rates, and recirculation rates vs. real-world performance after 2–3 years of operation; Early Bag Storage (EBS) sizing: how do high performers (FRA, AMS, ATL) actually scale EBS, and what does under-sized EBS cost in operational flexibility?; The integrated testing question: what does adequate end-to-end BHS commissioning look like, and what's the minimum schedule that doesn't compress it into theater?.

How This Report Was Produced

This report is the product of the Transform Airports AI Research Council — a multi-agent analytical system running on Anthropic's Claude models. Rather than a single model writing in one pass, the Council convenes a panel of specialized research agents, subjects their work to adversarial review, and verifies every claim before publication. The process runs in four stages, with human review at two checkpoints.

Stage 1 — Independent research. Each research agent below produced an evidence brief without seeing the others' work, so the analysis draws on genuinely independent lines of evidence rather than a single line of reasoning.

Stage 2 — Synthesis and adversarial review. A Strategist synthesized the briefs into an argument; a Red Team attacked it across two rounds; the Strategist revised after each.

Stage 3 — Edit and fact-check. An Editor tightened the prose. A Fact-checker verified every number against the research briefs, removing or flagging what it could not source.

Stage 4 — Production. The verified draft was formatted into this document.

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- **Strategist** — synthesized the research briefs into a single argumentative draft.
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- **Editor** — tightened the prose and removed filler without adding new content.
- **Fact-checker** — verified every numerical and attributed claim against the briefs.

Each agent is governed by a versioned specification — its analytical mandate, evidentiary standards, source discipline, and argumentative posture. The complete technical definitions for every agent on the Council are published and openly available at:

<https://github.com/TransformAirports/ai-council/tree/main/.claude/agents>

The Filter Nobody Removed: BHS Failure Is a Governance Problem, and MWAA's Capital Program Is the Place to Prove Otherwise

Council Run: Baggage Handling Best Practices **Stage 3 — Final Draft (post fact-check)** **Date:** June 3, 2026

I. Opening

Early on the morning of March 27, 2008, the first British Airways flight cleared the gate at Heathrow Terminal 5 and pushed back toward the runway. It carried no checked bags.

This is not the ordinary shorthand of an aviation disaster — "the bags didn't make it." The first aircraft departed T5 with zero bags in the hold because a software filter written to keep test messages from contaminating the live operating system during integrated trials had never been switched off when the trials ended. The filter that protected the system during testing was still protecting it on opening morning. When the first real bags arrived at the first real induction stations, the message system refused to acknowledge them. They were routed to manual sorting. Manual sorting overflowed within hours.

By the fifth day, 500 flights had been cancelled, 28,000 bags were stranded in temporary storage, and British Airways was publicly demanding £10 million from BAA. The Vanderlande system was rated at 12,000 bags per hour. It had been designed for twenty years. It had been built. It had been tested. The filter cost almost nothing to remove, and no one had removed it. The £4.3 billion building it was installed in lost five days because of an artifact from rehearsal.

T5 is the most-studied configuration management failure in commercial aviation, and it sits inside a pattern so consistent that treating it as coincidence has become a form of professional malpractice. Denver in 1995. Hong Kong's Chek Lap Kok in 1998. Kuala Lumpur the same year. Berlin Brandenburg in 2020. Four continents. Four vendors. Four owners. Three decades. The same failure mode in different uniforms.

These failures cost what they cost because of a structural change in the airport business that almost no one connects to baggage. Before 1978, an airline carried your bag from origin to destination on a single airplane it owned, at an airport whose schedule it set. The Airline Deregulation Act killed that model. Hub-and-spoke replaced it. A checked bag at Atlanta, Chicago, or Dallas in 1985 was no longer cargo on one aircraft. It was a unit of inventory passing through a sorting facility with a 45-minute minimum connect time, threaded against the bank-and-bank-out rhythm of a carrier that did not own the airport and could not run the system. A BHS failure at a hub in 2008 is not categorically the same event as a baggage failure in 1975. It is the failure of an industrial sorting facility on which forty inbound aircraft and forty outbound aircraft depend,

simultaneously, for the next ninety minutes. The cost of that failure is what makes the governance question urgent.

Every owner of every failed BHS since 1995 knew it was risky. None of them delayed.

II. Executive Summary

Eight claims, each grounded in the evidence below.

1. The failure taxonomy is short and stable. Across DEN '95, HKG '98, T5 '08, and BER '20, three causes recur: integrated end-to-end testing compressed or cancelled under schedule pressure; throughput and operational-readiness assumptions written to satisfy the business case rather than the engineering reality; and an owner with no institutional ability to push the opening date back. The pattern produced 16 months of delay at Denver, £102 million in damage at Chek Lap Kok, 28,000 stranded bags at T5, and tens of thousands of catalogued defects at BER. Different upstream causes; the same downstream failure of governance.
2. The hardware layer of baggage sortation is mature; the software stack is not. Munich T2's Independent Carrier System has run since 2003 at vendor-reported 99.9% availability — self-disclosed, unaudited. Changi T3 ran more than 1,000,000 test bags through 90-day ORAT and opened without incident. The architecture choices — tilt-tray, cross-belt, ICS, DCV — are well understood. What is not mature is the integration layer: vendor-managed control software, third-party messaging interfaces, the cybersecurity perimeter around them. Berlin Brandenburg lost days of operation in 2025 to a ransomware attack on Collins Aerospace, the airport's outsourced check-in and BHS software provider.
3. Vendor read rates and field read rates are different numbers, and confusing them is how capital planners get fooled. IATA documents barcode read rates in actual operations at 75–85% against vendor specifications of 95–99%. Amsterdam Schiphol's operational RFID read rate landed at 94.1% against a 99% target. Tracking is a data layer on top of the physical system. A bag tracked at 99.9% accuracy that is sitting in an undersized Early Bag Storage or circling a saturated makeup carousel is still a late bag.
4. CapEx dominates procurement decisions but does not dominate cost. ACRP Research Report 252 (2023) places initial purchase and installation at 15–20% of the 20-year BHS lifecycle cost; capital broadly defined at 30–40%. Operations, maintenance, energy, and labor consume the remaining 60–70%. Annual maintenance alone runs approximately 10% of initial construction cost. Procurement that optimizes the smallest fraction of the lifecycle bill while accepting risk to the largest is standard institutional behavior — and the behavior that produces the deferred-maintenance tails airports later pay a second capital cycle to cure.
5. Integrated testing is the precursor most consistently compressed in failures and most consistently honored in successes. Munich Airport advised Denver early that a system Munich had taken two years to build and six months of 24/7 testing could not be safely commissioned on Denver's schedule. The advice was ignored. At T5, a 2007

paper from the BHS project team acknowledged in advance that limited testing time would compromise integration. Logged and ignored. Changi T4 ran 100-plus trials with 4,000 participants and over a million test bags before its 2017 opening. The airports that ran the trials opened the way they planned. The airports that compressed them produced the case studies.

6. A 10-minute sortation outage generates approximately 6,000 unresolved bags (BEUMER published operational analysis). The difference between an airport that recovers in hours and one that recovers in days is not equipment. It is pre-staffed manual fallback routing, defined escalation authority for degraded-mode operations, makeup-area buffer sized to absorb the backlog, and a redundant secondary sortation path designed into the schematic at the start. The first three cost almost nothing in capital. The fourth is irreversible once the tunnel cross-sections are poured.
7. MWAA's debt trajectory makes Concourse E schedule pressure a structural risk, not a project-management worry. The funding stack: a \$9.4 billion capital program (2025–2039) — \$6.99 billion at IAD, \$2.39 billion at DCA — with approximately \$5.5 billion to be issued in new bonds through 2028. DSCR stood at 3.16x in 2024 and is projected to decline toward 1.3x by 2028 against Moody's 1.1x downgrade trigger. Debt per O&D enplanement is projected to rise from \$223 in 2024 to roughly \$400 by 2028. Moody's revised the U.S. airport sector outlook to negative in May 2025. United holds approximately 62% of IAD enplanements. Its certain exposure if opening slips — delay penalties, foregone revenue, network reschedule — is visible and immediate. Its probabilistic exposure if commissioning fails is not. The certain cost will dominate the probabilistic one. That is the mechanism by which United will press to open Concourse E on time even if the system signals it is not ready.
8. The capital decision is not whether to invest but how to govern the investment. Industry practice benchmarks BHS at 4–8% of total airport capital deployment depending on architecture. Applied to the \$6.99 billion IAD program, that yields an analyst-derived envelope of roughly \$280–560 million in BHS exposure — large enough to move credit metrics if it slips, large enough to lock IAD's connection performance for the next thirty years. The right discipline at procurement: an integrated commissioning protocol, an explicit stop/go gate above the COO level, schematic-phase architecture decisions made with operators in the room, a redundant sortation path written into the schematic-design RFP, and contractual testing standards immune to airline schedule pressure. Each is a design-phase decision. None is a recovery move at commissioning.

III. The Argument

The pattern is older than most of the people running it

Walk through the four canonical cases as the engineers themselves saw them at the time, and the structural similarity is hard to dismiss.

Denver hired BAE Automated Systems to deliver, in just over two years, the largest automated BHS ever attempted: 17 miles of track, 4,000 destination-coded vehicles, 100 networked computers, 5,000 photo eyes, 400 radio receivers, 56 barcode scanners. A delegation from Munich — whose own substantially smaller BHS had taken two years to build and six months of 24/7 testing — told the city the schedule was a fantasy. Denver pressed forward. Mid-project, scope expanded from a single United concourse to all twenty carriers on three concourses, without rebaselining schedule, budget, or risk. The airport opened sixteen months late in February 1995. GAO RCED-95-35BR documented roughly \$360 million in direct delay costs: \$230 million in operating deficits, \$37 million in foregone income, \$86 million for the backup conventional baggage system and modifications to the automated one, \$8.3 million in emergency bond issuance fees. The City of Denver carried \$1.1 million per day in debt service while the building sat empty. The automated system limped along for ten more years at \$1 million per month in maintenance — serving only United's outbound bags on one concourse — before United decommissioned it in 2005. Total spend across a system that never functioned to specification: \$520–600 million. United's spokesman Jeff Green explained the shutdown in the plain language carriers use when they have run out of patience: "Using this system simply doesn't allow us to provide the level of quality customer service that we want to provide." It was a sentence the system had earned ten years earlier.

Post-mortems on Denver are often misread as a technology cautionary tale — "automation was too ambitious" — but the contemporaneous record does not support that reading. The GAO documented, in flat prose, that "the city of Denver initially planned to open DIA with a conventional baggage system to meet the scheduled October 1993 opening date, but later decided to accept the risk associated with building an automated baggage system." No formal risk assessment accompanied the decision. The Munich engineers had told them how it would end.

Hong Kong's Chek Lap Kok opened July 6, 1998, on a calendar the political authorities had set and the engineering schedule could not catch. Within the first week, the integrated airport management system collapsed across baggage, flight information displays, aircraft parking, and cargo. HACTL shut its cargo facility on July 7 and the government temporarily reactivated the closed Kai Tak facility until August. Up to 10,000 pieces of checked baggage went to wrong destinations. The Hong Kong Financial Secretary estimated £102 million in economic damage. The Legislative Council inquiry concluded that the TAMS consortium had not completed commissioning before the politically locked opening. The systems worked individually. They had not been tested together under real operational load — because there was no time, because the calendar had decided that early July would happen no matter what the testing schedule said.

Kuala Lumpur opened five days before Hong Kong with the same pathology. Heathrow T5 opened ten years later, with the most experienced owner, the largest budget, and the longest planning window in the catalogue — twenty years from concept to ribbon. Construction delays at the building had compressed BA's familiarization program; on-site training was deferred by six weeks. Willie Walsh later told the House of Commons Transport Select Committee, in language unusually direct for an airline CEO under oath,

that "we made some mistakes and in particular had compromised on the testing regime as a consequence of delays in the building of the terminal." Staff arrived at gates they could not find in a building still partly under finish-out. The first BA flight pushed back with no bags. The message filter from testing had not been removed. By the end of the opening weekend, 28,000 bags were in storage. Direct damage to BA was estimated at £16 million; industry-side damage including BAA settlements and reputational costs was substantially higher. A 2007 paper by the T5 BHS project team had said the quiet part out loud before opening day: "the challenge of testing the integrated software is related not only to its size and complexity but also to the limited time that will be available to test the software in its actual environment." The warning was on the record. The testing time was still cut.

Berlin Brandenburg is the partial exception. The primary failure mode at BER was fire suppression — a smoke extraction system designed to draw smoke downward, against the physics of hot gas. Its designer, a draughtsman whose business cards claimed an engineering credential he did not hold, had been the lead consultant on the design. TÜV's 2014 inspection identified 75,000 individual defects; inspections through opening day in 2020 catalogued approximately 120,000. The airport opened October 31, 2020, nine years late, during a pandemic, at €6.5 billion against an initial €2.8 billion estimate. The BHS itself was tested with 15,000 trial bags before opening. The structural lesson is governance: a supervisory board stocked with politicians who had no mechanism to override sunk-cost escalation cannot, in practice, hold an opening when the project is in trouble. The board structure decided the opening cadence. The board structure had no engineering judgment to apply.

Diane Vaughan named the institutional pattern after Challenger: normalization of deviance. The O-ring engineers were on record before launch; the launch happened anyway. Each acceptance of an off-nominal signal makes the next acceptance easier, until the threshold has moved and the organization can no longer see that it has. Munich's warning to Denver, the 2007 T5 paper, the Hong Kong commissioning slip, the BER TÜV catalogue — warnings heard, logged, and absorbed. Upstream the cases differ: scope creep at Denver, testing compression at T5, system-of-systems cascade at Hong Kong, envelope failure at BER. Each needs a different intervention — contracting discipline, schedule architecture, integration architecture, oversight competence. They share a downstream pathway: an opening date the institution could not move, accountability that died at organizational boundaries, warning signs absorbed into the schedule once they had arrived enough times to feel normal.

The downstream gate — the authority to hold the opening date when the readiness review says hold — is the single lever that catches all four upstream failures at the same point. It is also the lever most consistently surrendered before it is needed.

The hardware is mature. The software and integration layers are not. If the technology were the failure mode, you would expect the industry's worst BHS performance to track its newest deployments. It does not.

Munich T2's Independent Carrier System has run since 2003. BEUMER's operational reporting describes twenty-two years of operation with the original carrier belts showing no measurable wear, 99.9% availability, and a mean time to repair of 8–10 minutes — figures operator- and vendor-self-disclosed, unaudited in public literature. Treated with caution, they still describe a system that scaled to over 20,000 bags per hour as passenger volume grew from 25 million to 41 million annually without redesign. Frankfurt's first-generation ICS has operated continuously since 1972. Changi T3 opened without first-week failure of consequence because the team ran more than a million test bags through 90-day ORAT. Changi T4 added emulation before any physical testing — running the control software faster than real-time to validate behavior under peak demand. Newark Terminal A in 2023 ran 30 trials over six months, with 3,000 test bags and 700 trial passengers, and opened without becoming a case study.

The available sortation architectures — tilt-tray, cross-belt, ICS, DCV — are each well understood, and the choice is irreversible once tunnel geometry is fixed in concrete. Tilt-tray achieves 5,000–6,000 bags per hour per loop with a closed-loop topology in which a segment failure can affect the entire loop. Cross-belt is gentler on fragile items but speed-limited at 2.5 m/s, capping at 4,000–4,500 bags per hour. ICS handles 90-degree vertical lifts that tilt-tray cannot — useful for the long below-grade runs IAD requires — but the tote is roughly 48 inches against a bare bag's 30, reducing EDS throughput by about 25% unless EDS capacity is sized up at programming. DCVs are useful at the long underground distances that defined Denver's concept and that also define IAD's concourse layout, but they are expensive per lane-mile and complex to maintain at scale. Each is a real engineering choice with real trade-offs. None is mysterious. The mechanical layer is not the variable that decides whether a \$300 million capital project ends as Munich or as Denver.

The software and integration layers are different, and the gap between the marketing layer and the field is where capital planners get fooled. RFID vendors and the airlines that have deployed them report 99.9% read accuracy. Delta's claim is credible at Delta's installation, where it has been audited internally. It is not a universal claim. Amsterdam Schiphol, running a substantial RFID trial, achieved 94.1% actual read rates against a 99% target. IATA's industry data places average barcode read rates in actual operations at 85%; some practitioners place the field average at 75%. Vendor specifications for the same scanners routinely claim 95–99%. Maryland Aviation Administration's PEGS Chapter 16.2 accepts a 5% misread rate during normal operations. At an 1,800-bag-per-hour induction line, 5% misread is 90 bags an hour routed to manual encoding — the labor capacity of multiple staffed workstations.

These gaps are the field's normal performance, and they are what competent BHS architecture is built to absorb. The mistake is asking RFID to do something it was never engineered to do: increase sortation capacity. Tracking is a data layer on top of the physical system. A bag tracked to four-nines accuracy sitting in an undersized EBS or circling a saturated makeup carousel is still a late bag. What the system does with the bags it loses track of is decided by buffer geometry and the degraded-mode protocol, not by the scanner reading the tag.

The newer risk has migrated from the iron to the code. The 2025 ransomware attack on Collins Aerospace took down outsourced check-in and BHS software at Berlin Brandenburg for days [UNVERIFIED — HUMAN REVIEW: the attack's reach to multiple European airports beyond BER is not confirmed in the Stage 1 evidence; verify scope before public publication]. This is a maturity gap in the software supply chain — and the failure mode the U.S. industry has done the least preparation for.

The lifecycle math says the procurement is upside down

ACRP Research Report 252 (2023) is the National Academies' total-cost-of-ownership framework for airport BHS. Its central finding is unambiguous and inconvenient: initial purchase and installation account for 15–20% of the 20-year lifecycle cost. Capital broadly defined — major mid-life replacements included — runs 30–40%. Operations, maintenance, energy, and labor consume the remaining 60–70%. Annual maintenance alone runs approximately 10% of initial construction cost. A \$200 million BHS carries a \$20 million annual maintenance obligation; over a 15-year conveyor lifespan, that is \$300 million in maintenance against \$200 million in construction.

Most airports do not procure BHS this way. The standard practice is to specify and select on installed cost — dollars per linear foot, dollars per induction point, dollars per CTX machine — because that is what the procurement file is built to compare. The OPEX consequence shows up in year three as a maintenance line that drifts upward, in year five as a jam rate climbing past one per 10,000 bags, and in year eight as a manual labor cost the ground handler invoices to the carrier rather than the airport. BHS failure costs migrate to the parties that did not specify the system.

Two consequences follow. First, Early Bag Storage is almost universally undersized at procurement because airports specify EBS for static capacity — bags held — and operators run it for flow. Frankfurt, Amsterdam, and Atlanta size EBS for the rate at which the system can intake bags and release them in batches to match a departure wave, not for the number of bags that can sit on a rack. The undersized EBS does not fail on a normal Tuesday. It fails on the morning a sortation loop is degraded and four delayed departure waves arrive at once with international inbound connect bags layered on top. That morning is not theoretical at Dulles. It is a regularly occurring stress condition.

Second, underfunded maintenance is not a small problem that compounds slowly. It is the mechanism that converts a 1-jam-per-10,000-bags system into a 1-jam-per-1,000-bags system. That is a tenfold degradation, and it lands not as a single dramatic failure but as a steady increase in manual intervention that absorbs ramp labor and eats schedule margin every push period. The airport's income statement does not see most of it. The carrier's mishandled bag rate does, and so does the gate scoreboard at every airline's operating center.

Three parties own the bag. No party owns the system.

The cleanest operational explanation for why governance fails after a BHS is commissioned is that no single party owns the operational mission. The airport owns the

asset. The airline uses it to move its bags between its aircraft. The ground handler — a third party contracted by the airline, often the lowest of multiple bidders, often turning over staff at high rates — operates the asset on the ramp, on the floor, at the bag room. Three parties. One bag. No single accountable executive whose performance review depends on the bag's on-time arrival at the right aircraft.

This is the gap the governance literature has been talking around for forty years. Munich T2 works in part because Lufthansa runs the BHS through Lufthansa Industry Solutions on a single integrated operational model. Changi works in part because the airport authority operates the entire common-use ramp and the handlers under direct contract. Most U.S. hubs do not work this way. The carrier subcontracts handling. The handler hires from a regional labor market with high turnover. The airport's maintenance crew responds to equipment failures. The airline's IT vendor manages BSM messaging. The CUSS kiosk vendor manages check-in. TSA owns the screening matrix. On the morning something fails, who decides the next move is genuinely unclear, and the time it takes to resolve that question is the time the secondary path saturates.

Buying a better BHS without solving the three-party ownership problem is buying a more expensive version of the same operational outcome. The solution is not capital; it is contract — the use-and-lease agreement, the airline operating agreement, the handler subcontract, and the readiness-gate authority. Airports that recover well from sortation outages have a ramp duty manager with pre-delegated authority to redirect labor across carriers without checking with three operating offices and an airport operations center. The authority exists on paper before the morning it is needed. It is exercised in drill before it is exercised under fire. The airports that recover badly are the ones where the authority gets debated in real time, on a conference bridge, while the makeup carousel saturates.

Integrated testing is the risk-management mechanism, and the schedule pressure is structural

Every documented failure in the modern record includes a specific moment when integrated testing was shortened, rescope, or deferred because the construction schedule had slipped and the opening date had not. Construction schedules at this scale always slip. The opening date does not, because by the time the slippage matters, bond covenants and airline move-in commitments and political ribbon-cuttings have made it immovable. Compression goes where institutional resistance is weakest, which is testing.

The empirical floor for adequate BHS commissioning is well documented. Changi T3 ran more than a million test bags through 90 days of ORAT. Changi T4 ran more than 100 trials with 2,500 airport staff and 1,500 volunteers. Newark Terminal A ran 30 trials over six months. Munich's T1 satellite ran 32 trial days over roughly two months. The IATA ORAT standard places the integrated operational trial at no less than four months before opening. A defensible floor for a system of IAD's complexity is six to nine months from iSAT completion to live operations. None of the airports that produced disaster case studies came close.

The deeper point is structural. The pattern is not a series of bad managers. It is the predictable product of a financing and accountability structure that puts schedule pressure on every project of this scale. Bond financing carries opening-date covenants. Airline move-in commitments precede system validation. Vendor performance specifications measure component throughput, not integrated system behavior. Commissioning, sitting at the end of the schedule, is the last line item and therefore the first to be cut when the schedule slips. The owner who delays incurs a certain, immediate, visible cost. The owner who does not incurs a probabilistic cost that, if it materializes, will be reported as the operator's mistake rather than the financier's. The arithmetic favors compression at the margin where it produces disasters. Where the institutional muscle to push back exists, the arithmetic does not win. Changi has it. Munich has it. Most U.S. hubs, governed by authorities with political boards and use-agreement carriers, do not.

Degraded-mode operations is the most under-designed part of the BHS

When a sortation loop fails — and at scale, it will — the question is whether the airport recovers in hours or in days. The difference is not equipment. It is whether the airport has pre-staffed manual fallback routing, a defined escalation path that does not require a control-room decision in real time, a makeup-area buffer sized to absorb redirected bags without re-saturating the secondary path, a communication protocol with carriers that prevents simultaneous check-in surges from re-saturating that route, and a redundant secondary sortation path designed into the schematic. BEUMER's published operational analysis estimates that a ten-minute sortation outage generates approximately 6,000 unresolved bags requiring manual intervention. An airport with pre-positioned staff and a defined fallback procedure can clear that backlog within a single bank window. An airport without can turn it into a multi-day cascade.

The first four elements require almost no capital expenditure. They require governance documents, drills, and a ramp duty manager with the authority — pre-delegated, in writing — to pull labor from normal positions into emergency sort positions without checking with three carriers and an airport operations center. The fifth — the redundant sortation path — is a schematic-design choice. It is also essentially free if designed in from the start, and prohibitively expensive if retrofitted. Most BHS designs optimize for nominal throughput and are never formally tested at degraded capacity. That is a schematic-design decision that becomes irreversible the moment the tunnel cross-sections are poured.

The read-rate metric is the polite way of avoiding the harder question of what happens to the bags the system loses track of. A 99.5% read rate with no degraded-mode protocol is operationally inferior to a 95% read rate with a practiced one. The first sounds better in a board presentation. The second wins on the morning that matters.

IV. The Counter-Case

The strongest argument against the governance thesis is not that the governance failures didn't happen. It is that the governance framing, applied to airports whose problem is something else, becomes a distraction. The case has weight and deserves to be stated at full strength.

The first counter-argument is physical. Every BHS has a hard ceiling set by its iron. No governance improvement, no testing discipline, no software layer expands that ceiling. At airports where peak-hour bag volumes are pressing against a 30-year-old system running at 90% utilization, the governance lessons from T5 and DEN are not the relevant ones. The relevant question is whether the pipe has enough capacity. Seattle-Tacoma is the contemporary reference: a system of 10 miles of conveyor, 3,000 motors, and 28 CTX machines, physically saturated at approximately 60 million annual passengers, being replaced in three phases over twelve years at approximately \$1 billion. That program is not governance remediation. It is a capacity response to a binding physical constraint, executed under live operations. Melbourne's \$500 million international BHS replacement — doubling capacity from 1,800 to 4,000+ bags per hour to support a planned doubling of international traffic from 12–13 million to 24–25 million annually — is the same logic. The system Melbourne is replacing is not failing because someone didn't test it. It is being replaced because it has run out of room.

The second counter-argument is asset life. Industry standard design life for BHS conveyors is approximately 15 years without major refurbishment. The U.S. is currently in a renewal cycle for systems installed in the late 1980s and 1990s — twice their design life. Predictive maintenance software can give a three-week warning that a bearing is failing. It cannot replace the bearing. It cannot increase the conveyor's rated speed. It cannot add sortation capacity the system was never designed to have. At the point of asset exhaustion, the choice is replace or close.

The third concerns EBS. Undersized EBS is sometimes framed as an operational problem solved by smarter scheduling. That framing is wrong: throughput rate is a physical specification driven by induction and retrieval bandwidth, square footage, and equipment capacity. The cleverest release algorithm on an undersized EBS will still lose bags when the wave breaks. Changi T4's EBS holds 4,000 bags; Changi T2's, 2,400; Schiphol's is sized for flow on a similar logic. Each was sized for the throughput its schedule requires. The fix is physical, and it belongs in the capital program, not the operations procedure.

The fourth extends the EBS point: physical redundancy is the only reliable resilience strategy. Software-managed degraded-mode operations require functioning physical infrastructure to switch between. If the schematic did not include a redundant sortation path, no amount of pre-delegated authority and rehearsed fallback will produce a non-degraded recovery. Governance is downstream of the capital decision that determined whether there is a fallback path at all. If you did not build the redundancy, you cannot govern your way to resilience.

The fifth is the global comparison the technology layer cannot explain on its own. IATA's 2024 mishandling data shows Asia-Pacific at 3.1 bags per 1,000 passengers, North and South America at 5.5, Europe at 12.3 — roughly four times the spread between Asia and Europe. Technologists attribute Asia's lead to RFID penetration (88% full adoption in China and North Asia against roughly 40% in Europe). But Asia also has the newest BHS infrastructure on the planet. Changi, Hamad, Incheon, KLIA, and a wave of Chinese greenfield airports were built to current specification within the last fifteen years. The technology advantage and the infrastructure advantage arrived together. Attributing the operational performance gap solely to software adoption flatters software vendors and undersells the role of having modern iron in the ground.

The sixth is the failure taxonomy itself. The four canonical cases are not four instances of the same failure. Denver was scope creep layered under testing compression. T5 was testing compression nested inside an organizational-boundary failure. Hong Kong was a cascading IT failure in which BHS was a downstream casualty. BER was an envelope failure of fire-life-safety and general construction in which the BHS appears as a victim. Lumping these into "governance failure" is convenient for a thesis. It smooths over the fact that the interventions required to prevent each one differ. Scope creep requires contracting discipline. Testing compression requires schedule architecture. Cascading IT requires system-of-systems engineering. Envelope failures require honest oversight boards. The thesis owes the reader a more precise taxonomy than "governance."

The seventh is the structural incentive problem. The carriers and vendors who participate in BHS commissioning have an incentive structure that runs against the owner's discipline. Vendors are paid against commissioning milestones; declaring the system "not ready" delays payment. Carriers have contracted move-in dates. Owners who delay face bond market consequences for the publicized slip. None of the parties at the readiness review has an unalloyed financial incentive to call the system not ready. Governance discipline in the abstract is not enough. It has to be built into the contracts before the pressure to compress arrives.

The eighth is regulatory and political reality. The owner does not control the full schedule. TSA's PGDS revision cycle, the iSAT clock that starts at 30% design approval, the 60-day TSA review window on facility modifications, the Buy American supply chain on AIP-funded components — all introduce schedule constraints the airport cannot relax. The PGDS itself moves: v8.0 is current; v9.0 is in development. A BHS designed today to v8.0 may need re-engineering to v9.0 before iSAT. The DCA perimeter rule moved only slightly in 2024 after two decades of stasis, when Congress added 10 new daily slot exemptions; the next slot expansion will arrive without consultation on BHS sizing at DCA. The political environment around MWAA's \$9 billion-plus capital program is the source of the schedule pressure that produces compression.

The ninth is the calibration warning the U.S. industry has failed to internalize: BHS sizing is a long bet on carrier behavior, and carriers leave. Pittsburgh International was built in 1992 to US Airways' hub specifications. US Airways dehubbed in 2004. The Allegheny County Airport Authority is now executing roughly \$1.5 billion in right-sizing

capital to undo the stranded infrastructure of a carrier's strategic withdrawal. United at IAD today is exactly the carrier concentration that created the Pittsburgh problem. Even an institutionally disciplined airport, building the right BHS at the right pace with the right testing, is making a thirty-year bet on one dominant carrier's network decisions. If that carrier rebases at year fifteen, the most beautifully governed BHS in the country is a stranded asset.

These objections deserve to be taken seriously. The next section does so.

V. Why the Counter-Case Is Insufficient

Some of those counter-arguments are conceded directly. Some are dissolved.

What is conceded. EBS sizing is conceded. EBS is a physical specification; throughput, footprint, and bandwidth are not amenable to operational improvement after the fact. Frankfurt and Amsterdam sized for flow because they had operators in the schematic-design room early. The fix happens at design, not at commissioning.

Physical redundancy is conceded. If the secondary loop is not in the concrete, no degraded-mode protocol recovers in hours. The redundant-path question is the most important schematic-phase decision for the IAD program, and it is currently inside the eighteen-month window where it can still be specified.

The carrier and vendor incentive objection is conceded without hesitation. At T5, BA had the most to gain from opening on schedule. At Denver, United had the most to gain from opening on schedule. At Concourse E in 2026, United will have the most to gain from opening on schedule. The owner who relies on the carrier's voluntary support for a testing delay is relying on the wrong party. Any MWAA BHS program should include a contractually defined testing-completion standard not subject to modification under airline pressure, signed by United in advance, with a stop/go gate above the COO level. The 2024–2039 use agreement is the natural place. The clause is not there yet.

Pittsburgh is conceded. The BHS sizing decision at IAD is a thirty-year bet on United's network strategy. The mitigation is not to refuse the bet — there is no version of the IAD program that does not size for United — but to design for carrier-agnostic flexibility: common-use matrix architecture, contractual reassignability if United's share declines, EBS sized for flow such that it serves a network of any shape. Pittsburgh had none of these.

The taxonomy objection is partly conceded. The four upstream failure modes are different and the report names them separately. They share a downstream pathway: an opening date the institution could not move, accountability that died at organizational boundaries, warnings logged and absorbed. "Governance" is shorthand for that pathway, not a denial of the upstream distinctions. The downstream gate is the leverage point precisely because the upstream causes are difficult to prevent project by project.

What is dissolved. The capacity-exhaustion argument is the strongest dissolved one because it is the most cleanly bounded. Seattle-Tacoma's BHS Optimization Program and Melbourne's international BHS replacement are real infrastructure responses to real physical constraints. They do not contradict the governance thesis; they bound it. Conditional on building physical infrastructure, governance discipline around testing, integration, and the opening date determines whether the new infrastructure performs at its design specification or repeats the cycle. The right question for Seattle was never "should we replace it" but "how do we replace it on a live airport without producing a Denver-shaped disaster on the morning of cutover." The answer involves twelve years of phased construction, dual-track operation during cutover, and an iSAT clock that started years before the program ended. Capacity and governance are not substitutes. They are sequential.

The aging-asset argument runs the same way. Systems at twice their design life must be replaced. That tells you nothing about what to build, how to size it, how to test it, or how to phase the cutover. Replacing aging assets well at a live hub is governance-intensive — night-work windows that compress productive hours, dual-track operation across cutover, TSA recertification that cannot be crashed, utility relocations under active aprons that require FAA-coordinated airside access windows. Replacing them well is harder than the capital request suggests.

The Asia-vs-Europe comparison is partly conceded and partly dissolved. The infrastructure-vintage advantage in Asia is real and partly causally independent of the governance advantage. The Asian hubs did not first build modern infrastructure and then layer governance on top; they built both in parallel, often because the same political authority that approved the capital also approved the governance regime. Attributing the spread solely to governance overclaims. Attributing it solely to infrastructure also overclaims. The U.S. industry is underweight on both.

The regulatory and political objection is correct on the facts and is exactly why the governance thesis matters more, not less. The PGDS revision cycle, the iSAT clock, and the Buy American provision cannot be moved within a project timeline. BHS design today must include margin against the announced direction of v9.0, and MWAA should engage TSA on draft v9.0 inside the program's design phase rather than learning the new standard when iSAT begins. The opening date itself is the one variable airport leadership entirely controls — if it chooses to. Every airport that failed had the authority to delay. None used it. The political environment makes that authority harder to use. It does not make it impossible. MWAA's 17-member board — appointed from three states and the federal government — is exactly the political condition in which the discipline must be architected before the pressure arrives. After it arrives, the procedural argument will be lost.

The framing is not wrong, but the politically and physically constrained environment in which it must operate makes it harder to apply than the framing alone suggests. The conclusion holds: governance discipline must be built into the contracts, the schematic design, the board procedure, and the construction sequence — not added at

commissioning. The airports that fail are not the ones whose program managers were incompetent. They are the ones whose institutional architecture did not protect the testing window when the pressure to compress it arrived.

VI. Implications for MWAA

The conditions that produce a Denver-shaped or T5-shaped outcome are not hypothetical at IAD. They are the operating environment of the next thirty months.

The schedule pressure is already in the bond documents. Concourse E is targeted for autumn 2026. The funding stack: a \$9.4 billion MWAA capital program (2025–2039) — \$6.99 billion at IAD, \$2.39 billion at DCA — with approximately \$5.5 billion in new bonds to be issued through 2028 and the balance carried by prior bonds, PFC, AIP, and BIL/ATP grants. Debt per O&D enplanement is projected to rise from \$223 in 2024 to approximately \$400 by 2028. Debt service coverage will decline from 3.16x in 2024 toward 1.3x by 2028 against Moody's downgrade trigger at 1.1x. Moody's changed the U.S. airport sector outlook to negative in May 2025. United, holding roughly 62% of IAD enplanements, has announced 280 daily departures by April 2025 and a \$500 million-plus Concourse E project sized for fourteen United gates and the elimination of mobile-lounge ground boarding. The IIJA Airport Terminal Program's FY2026 round is the final ATP announcement; AIP does not fund BHS; the PFC cap has been frozen at \$4.50 since 2000 and was not raised in the FAA Reauthorization Act of 2024. BHS scope not already inside an approved program will be financed at whatever spread the bond market quotes against a negative-outlook sector and a declining DSCR. The financial pressure that drove Denver's mayor to hold the February 1995 opening date will be applied to MWAA's board in 2026.

Protect the testing window contractually, because protecting it operationally will fail. The Munich-style minimum — two years of build, six months of 24/7 testing — was the original benchmark for a substantially smaller system. Changi's standards — a million test bags, 100+ trials, 4,000 participants — are the modern ones. IATA's minimum is four months of integrated operational trial before opening. Six to nine months from iSAT completion to live operation is a defensible floor for a system of IAD's complexity. The question is not whether ORAT will be planned. It will be. The question is whether it will be protected when the construction schedule slips. Construction schedules slip. The testing completion standard for Concourse E's BHS should be defined contractually at the highest level of MWAA's governance, signed by United, and structured so that the opening date adjusts to the testing readiness rather than the other way around. The 2024–2039 use agreement is the natural place. A useful template phrase: *No commissioning shall be deemed complete until the integrated operational trial has demonstrated, across no fewer than ninety contiguous days, the throughput, jam rate, and recovery-mode performance specified in Schedule [X]; the opening date shall be set by this standard and shall adjust without penalty to the Authority if the standard is not met.* That sentence does not exist anywhere in the current use agreement. It is the sentence that protects MWAA from itself.

Specify the architecture before the tunnel geometry is set. Tunnel cross-sections, makeup-room geometry, EBS footprint, and sortation architecture are decided in schematic design and locked in concrete by 60%. ICS deserves serious consideration at IAD given the long below-grade distances between the Main Terminal and the concourses — the same engineering condition that drove Denver to DCV in 1992. ICS handles the vertical lifts tilt-tray cannot. ICS energy consumption runs roughly a third of conventional conveyor for equivalent throughput. The 25% EDS-throughput reduction from the larger tote can be designed around by sizing EDS up at programming — but only if the specification is set against the tote dimension rather than the bare bag dimension. EDS programming should treat the published v9.0 direction as a margin assumption, not a future re-engineering surprise. If the tunnel is designed first and the BHS architecture is fit to whatever the tunnel will accommodate, IAD will own — for thirty years — a suboptimal system locked into a below-grade structure that cost a great deal to build. Cybersecurity — software stack architecture, isolation from third-party internet-facing systems, vendor concentration risk after the 2025 BER ransomware event — belongs in the same schematic-phase package, not as a vendor add-on after commissioning.

Write the redundant sortation path into the schematic-design RFP.

Conditional on the counter-case being conceded, this is the most important single recommendation in this report. If the redundant secondary sortation path is not specified in the RFP before tunnel cross-sections are fixed, no governance protocol can produce a non-degraded recovery from a sortation loop failure. The cost of designing a redundant path in at schematic is small. The cost of retrofitting one after the concrete is poured is, in the published experience of Seattle-Tacoma and Melbourne, ten to twenty times the design-phase figure [UNVERIFIED — HUMAN REVIEW: the specific 10–20× retrofit multiplier is a planning rule-of-thumb not directly cited in Stage 1 briefs; replace with sourced figure or remove before publication]. The Pittsburgh lesson sharpens the recommendation: a redundant sortation path that is carrier-agnostic — common-use matrix, contractual reassignability — is also the hedge against the dehubbing scenario the dominant-carrier configuration creates. Two arguments converge on the same physical specification.

Size EBS for flow, with the operators in the room. At a hub with United's wave structure and international transfer flows, EBS must be sized for flow rather than static capacity. Induct and retrieval bandwidth should be specified against the worst-case interaction of early international check-in, late-connecting domestic, and a degraded sortation loop. Frankfurt and Amsterdam are the references; Melbourne's new system, with a dynamic bag store of 1,400 bags supporting anytime check-in at a 24–25 million international target, is the most recent benchmark. The chronic failure at design is that EBS gets sized by engineers optimizing for the average operating day, with the operating chain — IAD's COO organization, United's ramp leadership, the ground handling subcontractor — absent or junior in the room. The conceptual error has a procedural remedy. It needs to be applied in the next eighteen months. A flow-sized EBS serves any network shape, not just the one the current carrier is operating.

Solve the three-party ownership gap in the use agreement. Picture the bag room at 5:15 a.m. on a winter morning at IAD, with United's first transatlantic bank pushing toward 5:55, the inbound from FRA already at the makeup carousel, and a fault indicator lighting on the primary sortation loop. The ramp duty manager on the floor needs to decide in the next two minutes whether to redirect labor from a Star Alliance partner's positions to United's, and the question of who has the authority to make that call is — in the current contractual posture — genuinely unclear. The Munich and Changi models are not available at IAD; the U.S. labor and contracting structure does not bend that far. What is available is a pre-delegated authority structure for ramp duty managers to redirect labor across carriers in degraded-mode events, written into the use agreement and the handler subcontract before commissioning, drilled before the first sortation outage, and not subject to a real-time conference bridge to invoke. The negotiation should be completed before commissioning, not after the first outage produces 6,000 bags that an industry analyst will quote in a quarterly review.

The constructability problem at a live hub is its own discipline. Concourse E is not a greenfield. It is incremental construction inside an operating international hub with United running 280 daily departures through it. The record is unsentimental: night-work windows that compress productive hours from twelve to four, dual-track BHS operation across cutover (which doubles the operating footprint while construction is underway), TSA recertification windows that cannot be crashed, utility relocations under active aprons that require FAA-coordinated airside access weeks in advance, iSAT clocks that start at 30% design approval and run against unstoppable calendar time. Each is a degree of freedom the project does not have. Each creates pressure that — if not absorbed in the schedule architecture — will be absorbed in the testing window. Seattle's twelve-year phasing is not exotic. It is what constructability discipline at a live hub looks like when it is taken seriously. Fast at a live hub is not a virtue.

The governance gate is the structural protection. MWAA's 17-member board, appointed across three states and the federal executive, is the structure in which the discipline to delay an opening date must be architected in advance. The CEO who walks into a Moody's call in late 2026 explaining that Concourse E is opening late because the system was not ready will be making a far easier defense than the CEO who walks in explaining that Concourse E opened on time and produced a T5-class commissioning failure. The first conversation is uncomfortable; the second is a credit event. The structural separation between the political authority that sets the date and the operational authority that approves the readiness must be written into board procedure, into the CEO's authority, and into the use agreement before the pressure arrives. After it arrives, the procedural argument will be lost.

Spend on contractual testing gates. Spend on schematic-phase architecture and EBS-flow decisions made with operators in the room. Spend on the redundant sortation path before the concrete is poured. Spend on pre-delegated degraded-mode authority and the drills that make it real. Spend on cybersecurity architecture as a schematic-design line item, not a post-commissioning vendor add-on. Spend on a board procedure that protects the operational stop/go decision from the political pressure that will be applied

to it. None of these is the largest capital line in the program. Each shifts where the program spends its discretion. A \$241.5 million Austin-Bergstrom-scale BHS at IAD, properly designed and commissioned, is a defensible figure for the system the program needs. A BHS at the upper end of the \$280–560 million envelope with deferred integrated testing and optimistic throughput assumptions is the Denver shape — not Denver's full ten-year cost, but the commissioning-event piece of it. The interesting variable is not the spread between the capital numbers. It is the testing schedule and the governance structure that decide whether each dollar buys reliable infrastructure or buys the next case study.

VII. Coda

There is a moment at every airport, on the first day of operations at a new facility, when the first bag rides the first belt for the first time. At Heathrow Terminal 5 in the early hours of March 27, 2008, that bag never reached its aircraft, because a software filter from rehearsal had not been removed, and because no one at the company that built the system or the carrier that occupied it or the airport that owned it was empowered to delay the morning long enough to find out. The filter cost almost nothing to remove. The direct loss to British Airways was £16 million, with industry-side damage including BAA settlements and reputational costs substantially larger — and a decade of institutional memory loss whose true cost is the next airport that opens with a filter still in place.

At Dulles, the morning that matters has not arrived yet. It is approximately eighteen months out. The schematic design package is still on the table. The use agreement can still be amended. The board procedure can still be written. The ramp duty manager who will be standing on the floor at 5 a.m. on opening day, holding a clipboard, with — or without — pre-delegated authority to redirect labor across three carriers, is already on the payroll. Whether the authority will be in writing by the time that morning arrives is a decision being made now, by people whose decision it is, in rooms that do not yet contain the operator who will inherit the consequence.

Stage 3 — Final draft after fact-check pass on Strategist v3 and Editor pass, June 3, 2026. Two passages tagged `[UNVERIFIED — HUMAN REVIEW]` for source confirmation prior to publication; see `outputs/stage3/fact-check-report.md` for the verification log.

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